higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE (VOCATIONAL)

## ELECTRONIC CONTROL AND DIGITAL ELECTRONICS
## NQF LEVEL 3

## 4 MARCH 2019

**This marking guideline consists of 6 pages.**

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## QUESTION 1

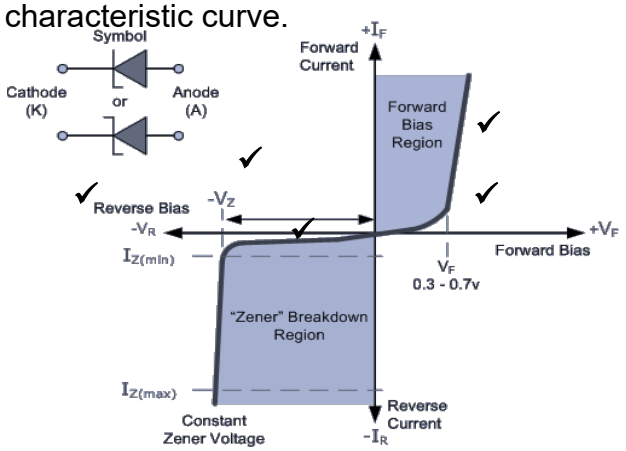
1.1
- Replace batteries when necessary.
- Remove batteries when storing meter for long periods.
- Keep instrument dry ... do not use in wet environments.
- Blow away dust after using in dusty areas.
- Clean with damp cloth and do not use chemicals or detergents when wiping.
- When workings on live circuits always make sure that the meter is in the correct range. (Any 5 × 1) (5)

1.2
1.2.1 D
1.2.2 A
1.2.3 E
1.2.4 B
1.2.5 C

(5 × 1) (5)

1.3 1.3.1
- Photo-diode operates in the reverse bias mode.
- An increase in light energy at the junction of the diode allows energised charge carriers to flow across the junction in the reverse direction.
- Alternatively, a photodiode is a light sensor that allows current to flow in one direction (reverse) when it absorbs photons (light). The lighter it absorbs, the greater the current. (Any 2 × 1) (2)

1.3.2 Zener diode characteristic curve.




(Give a mark for knee current in forward bias mode) (5)

1.4 1.4.1
- High current gain/amplification
- High voltage gain/amplification
- High power gain/amplification
- 180 degrees input to output phase shift
- High output impedance
- Average input impedance (Any 2 × 1) (2)

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1.4.2 DC current gain $\beta = \frac{IC}{IB}$

$= \frac{2 \times 10^{-3}}{0.04 \times 10^{-3}}$

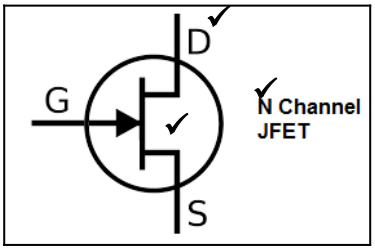
$= 50$ (3)

1.4.3 $\mathbf{h_{FE}}$ = DC current gain (1)

1.5 1.5.1
- A standard transformer allows AC to change direction of current at the secondary.
- The first half cycle will pass through D1, the forward biased diode, run through the load and the return to the transformer.
- The second half cycle does not pass the reverse biased diode D1, thus current does not pass through the load.
- Pulsating DC is created with only half cycles. (Any 3 × 1) (3)

1.5.2
- An inductor in connected in series with the load.
- The capacitor is connected in parallel with the load. (2)

1.6 1.6.1




(3)

1.6.2 The gate lead of the MOSFET is insulated from the channel by a thin layer of metal oxide. The gate is not part of the internal structure. (2)

1.7 1.7.1 Non-inverting op-amp (1)

1.7.2 Rf is used for negative feedback (1)

1.7.3
- Input and output signals are in phase
- High input impedance.
- The output has larger amplitude than the input signal. amplified.
- Negative feedback is used to stabilise the output. (Any 2 × 1) (2)

1.7.4 $Vout = \left(1 + \frac{Rf}{Rin}\right) . Vin$

$= \left(1 + \frac{20\,k\Omega}{2\,k\Omega}\right) . 0,5\ V$

$= 5,5\ V$ (3)

**[40]**

## QUESTION 2

2.1 2.1.1 $59_{10}$ to hexa

59 ÷ 16 = 3 remainder 0,6875 × 16 = 11 = B
3 ÷ 16 = 0 remainder 0,.1875 × 16 = 3
Answer = $3B_{16}$

2.1.2 $135_8$ to decimal

$1 \times 8^2 = 64$
$3 \times 8^1 = 24$
$5 \times 8° = 5$
$+93$

Answer = $93_{10}$

2.1.3 $11101110_2$ to octal

(011) (101) (110)
3 5 6
Answer = $356_8$

(3 × 3) (9)

2.2 Multiply $10101_2 \times 101_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 1 | 0 | 1 | 0 | 1 |
| | x | | | 1 | 0 | 1 |
| . | . | 1 | 0 | 1 | 0 | 1 |
| . | 0 | 0 | 0 | 0 | 0 | . |
| 1 | 0 | 1 | 0 | 1 | . | . |
| 1 | 1 | 0 | 1 | 0 | 0 | 1 |

Answer = $1101001_2$ (3)

2.3 Divide $10011_2 \div 110_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | 1 | 1 | ,1 |
| 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 0 |
| | | | - | 1 | 1 | 0 | | |
| | | | 0 | 1 | 0 | 1 | 1 | |
| | | | | - | 1 | 1 | 0 | |
| | | | | | 1 | 0 | 1 | 0 |
| | | | | | - | 1 | 1 | 0 |
| | | | | | | 1 | 0 | 0 |

Answer = $11,1_2$ (3)

2.4 Subtract $10100_2 - 101_2$

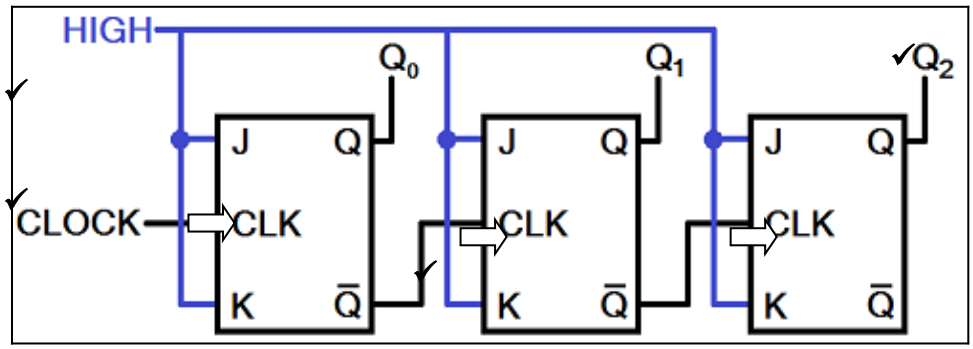
| | | | | |
|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 0 |
| | - | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 | 1 |

Answer = $1111_2$ (3)

2.5 2.5.1

- The decade counter has only ten states in its sequence.
- It counts from 0 to 9
- On the 11th count 1010, all flip flops reset.
- The count goes back to 0 and cycles again. (Any 3 × 1) (3)

2.5.2




**NOTE:** Sequence of clock can be drawn in opposite direction
Consider all similar and correct diagrams. (4)
**[25]**

**QUESTION 3**

3.1

- Tower
- Monitor/Screen
- Keyboard
- Mouse
- Printer/Scanner/External hard drive (Any 4 × 1) (4)

3.2 IDE slot/ATX power slot/RAM slot/PCI (peripheral or expansion) slot/CPU socket/Fan slot/Serta slot/System fan/BAT (3)

3.3

- The PS-2 or USB plug might not be seated or damaged
- The cable of the mouse might be damaged
- The optical pickup might be obstructed with dust
- Physical damage (Any 3 × 1) (3)

3.4

- Open the tower casing and remove all cables plugged onto motherboard.
- Remove the mounting screws that hold the board to the earthed chassis.
- Ensure that the new motherboard match the specifics the suit your needs.
- Install the CPU and cooling fan.
- Install memory modules and expansion cards.
- Connect all cables and wires to respected slots. (Any 5 × 1) (5)

**[15]**

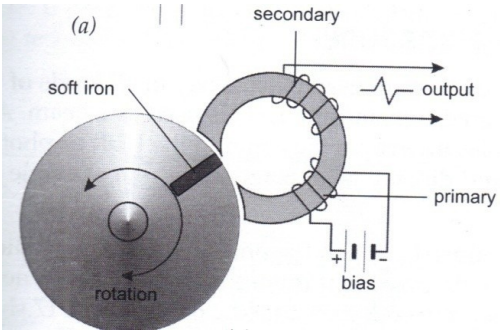
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## QUESTION 4

4.1
- The type of supply needed or available
- The quantities or values being measured.
- The accuracy needed for the application.
- The cost and availability of devices.
- The environment must suit the device operation.

(Accept: Accuracy/Stable/Reliability/Reproducible/Sensitivity) (5)

4.2
- Two coils are wound on the same core, allowing an induction of a magnetic field through the core.
- A soft iron or magnetic slug is attached to the rotating wheel.
- As the wheel turns, the slug passes the core of the windings and creates pulses.
- The pulses are counted and give an indication as to the speed or velocity of the wheel. (Any 3 × 1) (3)

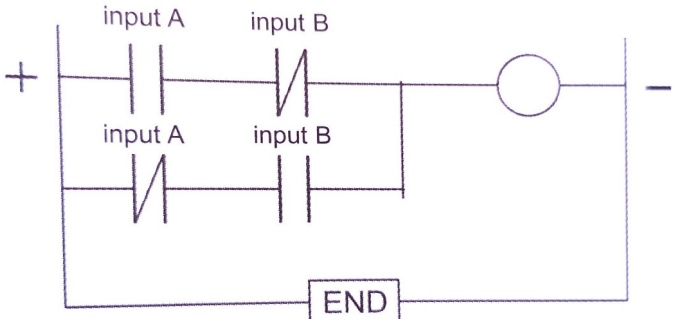



(2)
**[10]**

## QUESTION 5

5.1
- PLCs are very fast operating systems.
- Very compact size uses less space.
- Easy to change instructions using computer program during repair or maintenance.
- Easy to design operation using program, no wiring of relays needed.
- Less mechanical operational faults. (Any 4 × 1) (4)

5.2




- (1 mark for each symbol × 6) (6)

**[10]**

**TOTAL: 100**

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